Class 35: Project Requirements and Development Plan

Session 35 of 36 – MLOps Block

Data-driven Systems Engineering

18 December

Agenda

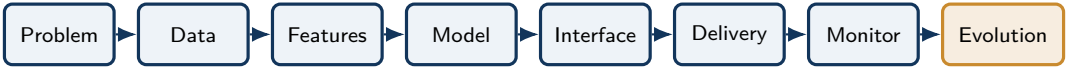
- From pitch to requirements: reconnecting to Class 25's practice
- Deliverables mapped to the full lifecycle
- Work Breakdown Structure and milestone sequencing
- Sprint and timeline thinking
- Definition of Ready and Definition of Done
- Risk register for ML projects

Planning is part of the engineering work

Learning goals

- Turn a scoped project pitch into a milestone sequence and Work Breakdown Structure.
- Map deliverables onto the full lifecycle: data, model, interface, delivery, and monitoring.
- Write Definition of Ready and Definition of Done criteria the team will actually use.
- Build a lightweight risk register with concrete responses.

Course Map and Running Cases



Today in the course

Focus: Project planning as lifecycle design.

Bridge: Class 33's pitch becomes a real project only once it has milestones, deliverables, and a way to tell whether work is ready or done; Class 34's guest session is a reminder that real engagements run on exactly this kind of plan.

Fraud case

In the fraud case, planning means sequencing data work, baseline modeling, service exposure, deployment checks, and monitoring criteria.

Pasteurization case

In the pasteurization case, planning means sequencing sensor understanding, forecasting or alerting logic, dashboard integration, and drift response design.

Course thread: The course thread is that a coherent project plan should already reflect the full lifecycle, not only the modeling step.

From your Class 33 pitch to this plan

- Your MVP becomes the first few milestones; your stretch goal becomes a later, optional milestone.
- Your two named risks become the first two rows of today's risk register.
- Your success criterion becomes the definition of done for the final milestone.

Milestones and WBS

- Proposal approval
- Data pipeline MVP
- Baseline model
- Deployable service or dashboard
- Monitoring active
- Final demonstration

WBS reminder

Break work into outputs that can be reviewed, not vague activity labels.

The WBS is about outputs, not vague activity

- Good work packages end in something reviewable: cleaned dataset, baseline notebook, API endpoint, dashboard page, monitoring report.
- The 100% rule helps ensure the WBS covers the whole scope without duplication.
- If a task cannot be evaluated by an external reviewer, it is often too vague.

Suggested semester deliverables

1. Data understanding and requirement brief.
2. Baseline model and evaluation report.
3. Packaged service or dashboard.
4. CI/CD or reproducibility automation.
5. Monitoring or drift analysis plan.

Deliverables should map to the lifecycle

Deliverable	Why it matters
Data pipeline artifact	Makes inputs reproducible and inspectable.
Baseline model	Creates a reference point for later improvement.
Deployable interface	Proves the model can be consumed by users or systems.
Automation artifact	Reduces manual, unrepeatable work.
Monitoring plan	Shows how the system remains trustworthy after release.

Sprint and timeline thinking

- Sprint 0: environment, repo structure, and scope.
- Sprint 1: data understanding and requirements.
- Sprint 2: baseline modeling.
- Sprint 3: serving and integration.
- Sprint 4: monitoring and final hardening.

Milestones reduce risk when they represent decisions

- Proposal approval means the team has a coherent scope.
- Data pipeline MVP means later modeling is grounded in real inputs.
- Baseline model means performance can be compared instead of guessed.
- Deployable service means integration risk is no longer abstract.
- Monitoring active means the project is thinking beyond the demo.

DoR and DoD for student teams

Definition of Ready

- Requirement is written
- Data source identified
- Acceptance criteria drafted

Definition of Done

- Code runs reproducibly
- Result is documented
- Test or demo evidence exists

A lightweight risk register helps project control

Risk	Example response
Data access or quality risk	Define fallback datasets and validation checks.
Model performance risk	Commit to a baseline and threshold review.
Integration risk	Deliver an API or dashboard slice early.
Schedule risk	Separate must-have deliverables from stretch goals.
Governance risk	Document assumptions, versions, and monitoring triggers.

Project framing documents already in this repo

Repo activity

- ‘documents/FraudDetectionAsaService.pdf‘ and ‘documents/FraudDetectionAsaService_presentation.pdf‘ show a fully written project framing for the fraud case.
- ‘documents/Forecasting_and_Drift_Pasteurization.pdf‘ shows the same for the pasteurization case.
- Use their structure, not their content, as a model for your own plan if your project follows a different idea.

Discussion prompt

Question

Which single milestone in your plan is actually a decision point, where the project could be redirected or stopped if things go wrong?

- This distinguishes a real milestone from a calendar date with a label on it.
- It also connects planning back to the risk register: a decision point is where a risk gets checked, not just hoped away.

Papers and materials

Papers and materials

Sculley et al. (2015), Hidden Technical Debt; Designing Machine Learning Systems; Project planning with epics and stories; Critical path and schedule planning basics.

From This Class to the Next

What stays with us

- A strong plan defines outputs, milestones as decision points, and named risks with responses.
- Milestones should demonstrate engineering progress, not only analytical activity.
- Good project planning anticipates delivery, monitoring, and change from the start.

Project use

This plan, together with the Class 33 pitch, is what each team brings into the proposal presentations.

Next connection

Class 36 covers the logistics and rubric for the final proposal-presentation session.

Course thread: Planning is complete only once it can be defended in front of reviewers, which is exactly what the next two sessions are for.